THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 768 cm²	$l+b=64$, $l=3b$: $b=16$, $l=48$
2	**(d)** 128 cm²	Height $=\sqrt{10^2-6^2}=8$ cm; area $=\frac{1}{2}(10+22)\times 8$
3	**(d)** 1080	$1000\times 120/100\times 90/100$
4	**(c)** 15	$2A/16$
5	**(d)** 840 cm²	Side $=29$; other half-diagonal $=21$; area $=\frac{1}{2}\times 40\times 42$
6	**(a)** 1232 m²	$(52-8)(36-8)$
7	**(d)** 924 cm²	$33^2+56^2=65^2$, so it is right-angled; area $=\frac{1}{2}\times 33\times 56$
8	**(b)** 405 cm²	$l+b=54$, $l=5b$: $b=9$, $l=45$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$12\times 5=60$ cm^2; R is the formula used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A is true by $\frac{1}{2}\times 10\times 8=40$; R is also true but is a separate fact, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$l+b=50$, $l-b=18$ (1); $l=34$, $b=16$, area $=544$ cm² (1).	1+1
12	New area $=100\times 130/100\times 80/100=104$ (1); change $=4\%$ (1).	1+1
13	$h=2A/b=15$ cm (1); new area $=270$ cm² (1).	1+1
14	$l+b=60$, $l-b=22$ (1); $l=41$, $b=19$, area $=779$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$b=20$ m, $l=60$ m (1); area $=1200$ m² (1); fencing cost $=160\times 15=$ ₹ 2400 (1).	1+1+1
16	Height $=\sqrt{15^2-9^2}=12$ cm (2); area $=\frac{1}{2}\times(14+32)\times 12=276$ cm² (1).	2+1
17	Garden − pond $=1200-80=1120$ m² (1); path $=(40+4)(30+4)-1200=296$ m² (2).	1+2
	OR: $b=18$ m, $l=72$ m (1); area $=1296$ m² (1); fencing cost $=180\times 12=$ ₹ 2160 (1).	1+1+1
18	Height $=\sqrt{20^2-12^2}=16$ cm (2); area $=\frac{1}{2}\times(20+44)\times 16=512$ cm² (1).	2+1
	OR: Garden − pond $=1750-108=1642$ m² (1); path $=(50+6)(35+6)-1750=546$ m² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{2}(50+30)\times 20=800$ m² (1). (ii) $800\times 15=$ ₹12,000 (1). (iii) $800\div 4=200$ mats (1). (iv) New area $=\frac{1}{2}(60+40)\times 20=1000$ m² (1); increase $=200$ m² (1).	1+1+1+2
20	Other diagonal $=2\times 384/24=32$ cm (2); side $=\sqrt{12^2+16^2}=20$ cm (1); altitude $=384/20=19.2$ cm (2).	2+1+2
	OR: Trapezium $=\frac{1}{2}(18+30)\times 8=192$ cm² (2); parallelogram $=144$ cm² (1); triangle $=\frac{1}{2}\times 12\times 8=48$ cm² (2).	2+1+2

Mistakes to avoid:

- Using the slant side instead of the perpendicular height in the area of a parallelogram or triangle.
- Forgetting the factor $\frac{1}{2}$ in the area of a triangle, trapezium or rhombus.
- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 12	$2A/30$
2	**(c)** 840 cm²	Side $= 37$; other half-diagonal $= 35$; area $= \frac{1}{2} \times 24 \times 70$
3	**(d)** 1620 m²	$(64-10)(40-10)$
4	**(c)** 1560 cm²	$39^2 + 80^2 = 89^2$, so it is right-angled; area $= \frac{1}{2} \times 39 \times 80$
5	**(a)** 900 cm²	$l + b = 75$, $l = 4b$: $b = 15$, $l = 60$
6	**(a)** 330 cm²	Height $= \sqrt{17^2 - 8^2} = 15$ cm; area $= \frac{1}{2}(14+30) \times 15$
7	**(d)** 1320	$1500 \times 110/100 \times 80/100$
8	**(c)** 12	$2A/35$
9	**(c)** A is true but R is false.	$\frac{1}{2}(8+12) \times 5 = 50$, so A is true. R omits the factor $\frac{1}{2}$, so it is false.
10	**(d)** A is false but R is true.	They have equal areas but need not be congruent, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New area $= 100 \times 140/100 \times 75/100 = 105$ (1); change $= 5\%$ (1).	1+1
12	$h = 2A/b = 14$ cm (1); new area $= 420$ cm² (1).	1+1
13	$l + b = 80$, $l - b = 24$ (1); $l = 52$, $b = 28$, area $= 1456$ cm² (1).	1+1
14	New area $= 100 \times 120/100 \times 70/100 = 84$ (1); change $= -16\%$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$b = 15$ m, $l = 75$ m (1); area $= 1125$ m² (1); fencing cost $= 180 \times 20 =$ ₹ 3600 (1).	1+1+1
16	Height $= \sqrt{25^2 - 7^2} = 24$ cm (2); area $= \frac{1}{2} \times (16+30) \times 24 = 552$ cm² (1).	2+1
17	Garden − pond $= 2400 - 150 = 2250$ m² (1); path $= (60+4)(40+4) - 2400 = 416$ m² (2).	1+2
	OR: $b = 21$ m, $l = 84$ m (1); area $= 1764$ m² (1); fencing cost $= 210 \times 10 =$ ₹ 2100 (1).	1+1+1
18	Height $= \sqrt{17^2 - 8^2} = 15$ cm (2); area $= \frac{1}{2} \times (18+34) \times 15 = 390$ cm² (1).	2+1
	OR: Garden − pond $= 1008 - 54 = 954$ m² (1); path $= (36+2)(28+2) - 1008 = 132$ m² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $24 \times 15 = 360$ m² (1). (ii) $\frac{1}{2} \times 24 \times 15 = 180$ m² (1). (iii) $360 \div 18 = 20$ m (1). (iv) Lawn: $360 \times 40 =$ ₹14,400 (1); half: ₹7,200 (1).	1+1+1+2
20	Other diagonal $= 2 \times 24/6 = 8$ cm (2); side $= \sqrt{3^2 + 4^2} = 5$ cm (1); altitude $= 24/5 = 4.8$ cm (2).	2+1+2
	OR: Trapezium $= \frac{1}{2}(12+28) \times 9 = 180$ cm² (2); parallelogram $= 108$ cm² (1); triangle $= \frac{1}{2} \times 16 \times 9 = 72$ cm² (2).	2+1+2

Mistakes to avoid:

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- Forgetting the factor $\frac{1}{2}$ in the area of a triangle, trapezium or rhombus.
- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm², or converting m² to cm² by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1320 cm²	$48^2 + 55^2 = 73^2$, so it is right-angled; area $= \frac{1}{2} \times 48 \times 55$
2	**(a)** 784 cm²	$l + b = 70$, $l = 4b$: $b = 14$, $l = 56$
3	**(d)** 320 cm²	Height $= \sqrt{20^2 - 12^2} = 16$ cm; area $= \frac{1}{2}(8 + 32) \times 16$
4	**(b)** 900	$800 \times 150/100 \times 75/100$
5	**(b)** 15	$2A/20$
6	**(a)** 2880 cm²	Side $= 82$; other half-diagonal $= 80$; area $= \frac{1}{2} \times 36 \times 160$
7	**(d)** 2812 m²	$(80 - 6)(44 - 6)$
8	**(c)** 1386 cm²	$36^2 + 77^2 = 85^2$, so it is right-angled; area $= \frac{1}{2} \times 36 \times 77$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{1}{2} \times 10 \times 24 = 120$; R is the formula used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A is true ($9^2 = 81$); R is true but unrelated to the area of a square.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$h = 2A/b = 16$ cm (1); new area $= 352$ cm² (1).	1+1
12	$l + b = 100$, $l - b = 30$ (1); $l = 65$, $b = 35$, area $= 2275$ cm² (1).	1+1
13	New area $= 100 \times 150/100 \times 60/100 = 90$ (1); change $= -10\%$ (1).	1+1
14	$h = 2A/b = 15$ cm (1); new area $= 510$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$b = 18$ m, $l = 54$ m (1); area $= 972$ m² (1); fencing cost $= 144 \times 25 =$ ₹ 3600 (1).	1+1+1
16	Height $= \sqrt{26^2 - 10^2} = 24$ cm (2); area $= \frac{1}{2} \times (22 + 42) \times 24 = 768$ cm² (1).	2+1
17	Garden − pond $= 1440 - 100 = 1340$ m² (1); path $= (45 + 4)(32 + 4) - 1440 = 324$ m² (2).	1+2
	OR: $b = 18$ m, $l = 90$ m (1); area $= 1620$ m² (1); fencing cost $= 216 \times 8 =$ ₹ 1728 (1).	1+1+1
18	Height $= \sqrt{25^2 - 15^2} = 20$ cm (2); area $= \frac{1}{2} \times (30 + 60) \times 20 = 900$ cm² (1).	2+1
	OR: Garden − pond $= 3500 - 240 = 3260$ m² (1); path $= (70 + 6)(50 + 6) - 3500 = 756$ m² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $12 \times 8 = 96$ m² (1). (ii) $5 \times 4 = 20$ m² (1). (iii) $96 - 20 = 76$ m² (1). (iv) $76 \times 250 =$ ₹19,000 (1+1).	1+1+1+2
20	Other diagonal $= 2 \times 864/36 = 48$ cm (2); side $= \sqrt{18^2 + 24^2} = 30$ cm (1); altitude $= 864/30 = 28.8$ cm (2).	2+1+2
	OR: Trapezium $= \frac{1}{2}(22 + 40) \times 14 = 434$ cm² (2); parallelogram $= 308$ cm² (1); triangle $= \frac{1}{2} \times 18 \times 14 = 126$ cm² (2).	2+1+2

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